

Project Quad

CUSTOM ESP32 FLIGHT CONTROLLER & PVC DRONE BUILD

● SCOPE

Combine a bare-metal ESP32 flight controller, custom sensor fusion, C++ control firmware, and hand-built PVC airframe hardware.

● STATE VECTOR

$x = [\varphi, \theta, \psi, \dot{\varphi}, \dot{\theta}, \dot{\psi}]^T$, representing attitude and angular rates.

● CORE TIMING

250 HZ LOOP

LOOP_TIME_US = 4000, producing a deterministic 250 Hz control loop.



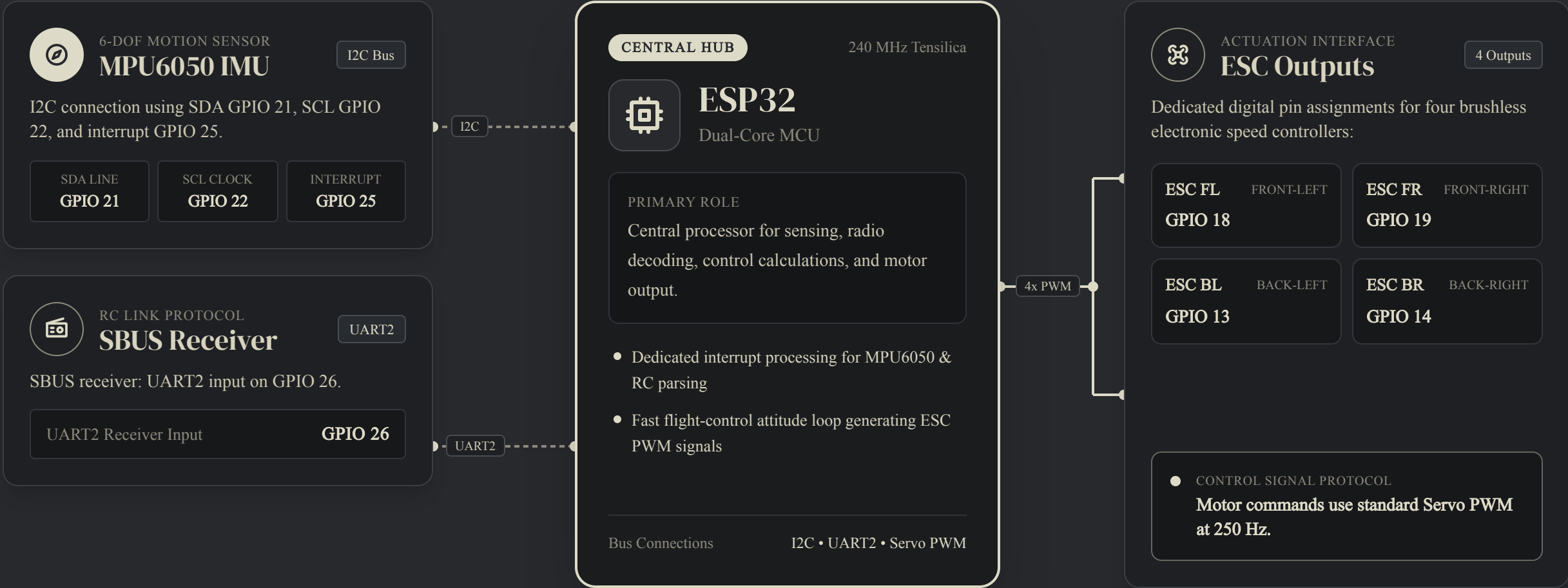
FIRMWARE_CONFIG.H

```
#define FIRMWARE_VERSION "V69.4-CINEMATIC"
```

```
#define LOOP_TIME_US 4000
```

Embedded Hardware & Physical Schematic

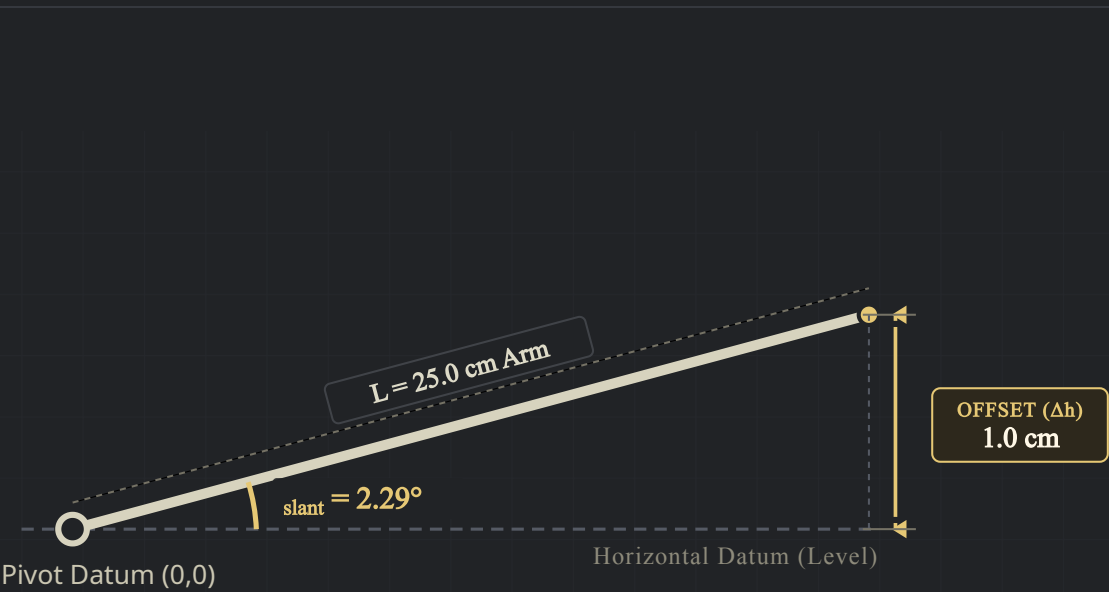
ESP32 → MPU6050, SBUS receiver, ESC FL, ESC FR, ESC BL, ESC BR



Frame Geometry & Mechanical Slant Analysis

● SIDE PROFILE GEOMETRY DIAGRAM

SCALE: EXAGGERATED 3× FOR VISIBILITY



⚠ Hand-built PVC frames introduce dimensional tolerances before software compensation begins. A 1.0 cm vertical offset across a 25.0 cm arm creates an inherent baseline tilt.

ANGULAR CALCULATION

TRIGONOMETRIC FORMULATION

SLANT ANGLE FORMULA

Slant angle: $\theta_{\text{slant}} = \arctan(1.0 / 25.0) \approx 2.29^\circ$

Bench measurements should populate `BaseRollOffset` and `BasePitchOffset` so the controller distinguishes frame geometry from actual motion.

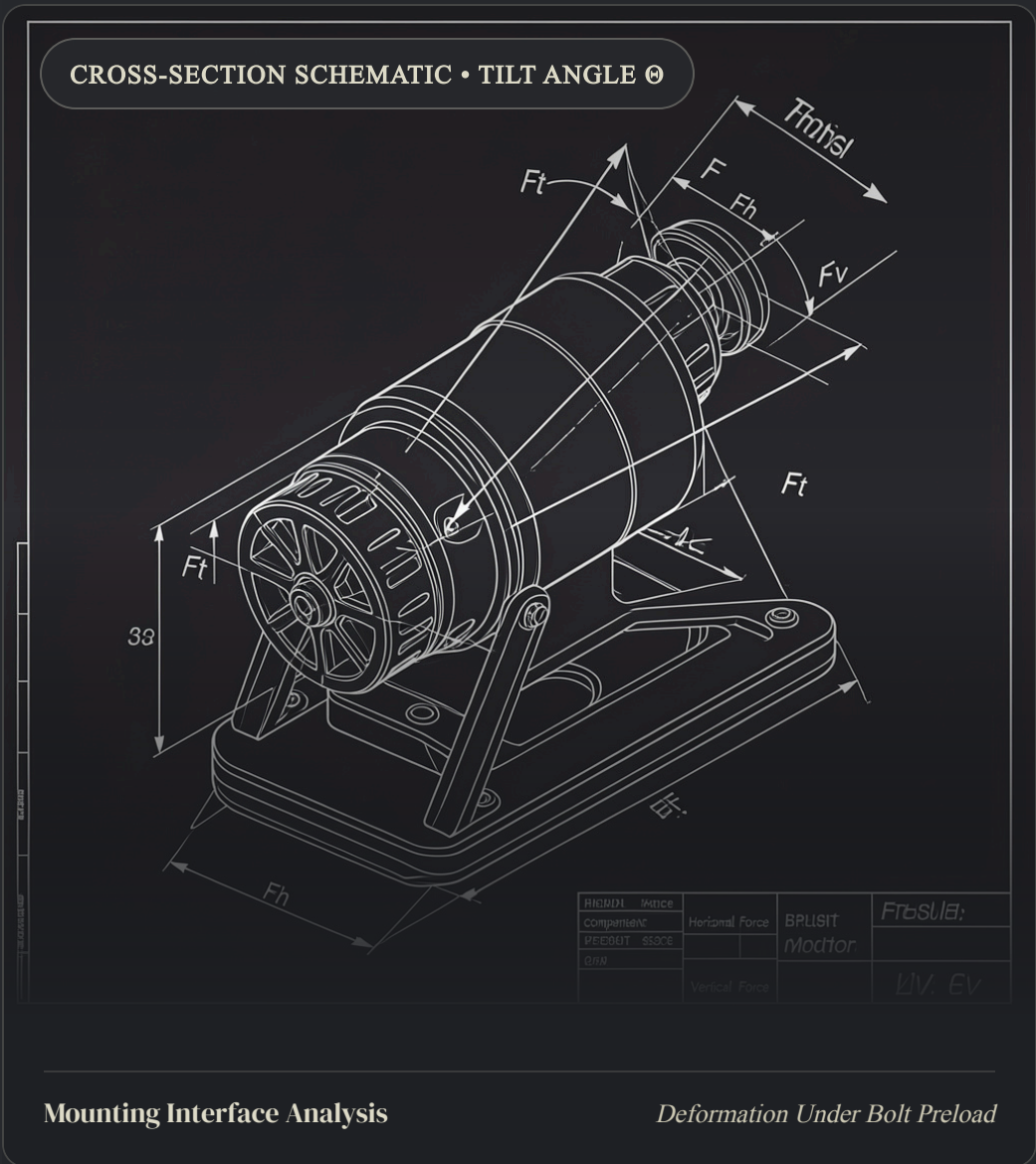
FIRMWARE CALIBRATION

C++

```
// Calibration offsets initialized in firmware  
CALIBRATION.CPP  
  
float BaseRollOffset = 0.0;  
float BasePitchOffset = 0.0;
```

Hardware Tare Reference:

SLANT CORRECTION = 2.29°



Tightened motor bolts can crush soft 3D-printed PLA under heat and screw tension.

A motor tilted only 1.0° diverts part of its thrust horizontally.

VECTOR FORCE DECOMPOSITION

Horizontal component: $F_h = F_t \sin(\theta)$; vertical component: $F_v = F_t \cos(\theta)$.

The resulting torque can make the aircraft behave like a pinwheel, forcing the yaw controller into continuous corrective effort.

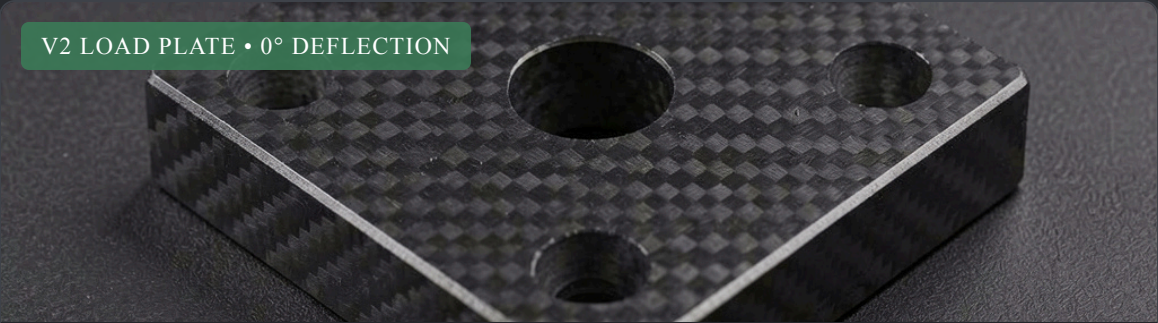
C++ YAW RATE CONTROLLER

Closed-Loop Correction

ErrorRateYaw = DesiredRateYaw - RateYaw;

PtermYaw = PRateYaw * ErrorRateYaw;

Iterative Hardware Solution: V1 vs. V2 Carbon Load Plates



V1: Narrow Footprint Contact

High localized pressure & plastic creep

V2: Wide "Snowshoe" Load Plate

Even tension dissipation & planar stability

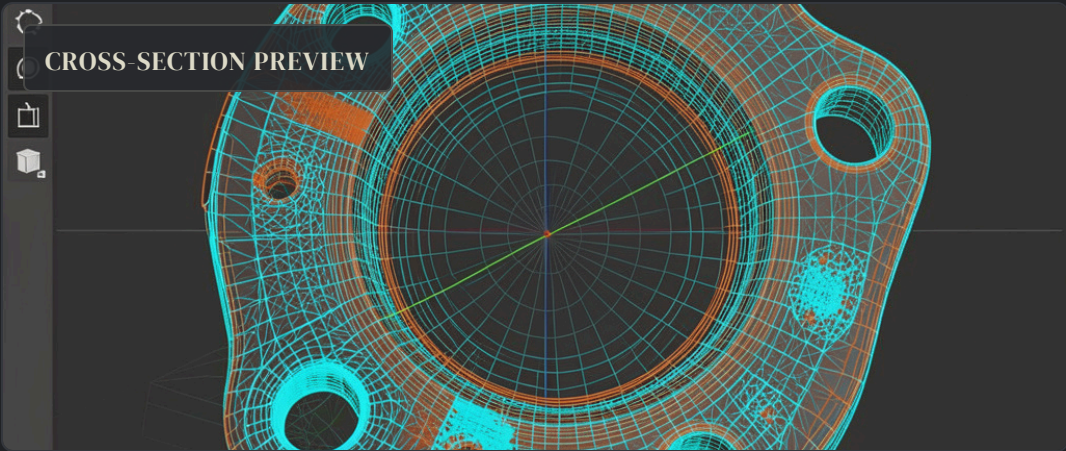
VERSION	GEOMETRY	RESULT
V1	1.6 mm thick, narrow footprint	Crushed plastic and sustained 1° motor tilt
V2	3.0 mm thick, 4 mm oversized margin	Distributed screw tension and 0° deflection

- The key improvement was not only thickness, but the larger “snowshoe” contact area.

- Load distribution preserves motor alignment under tightening and thermal stress.

Slicer Optimization for Structural Drone Parts

Decorative parts can use sparse infill, but structural motor hardware experiences concentrated screw and vibration loads.

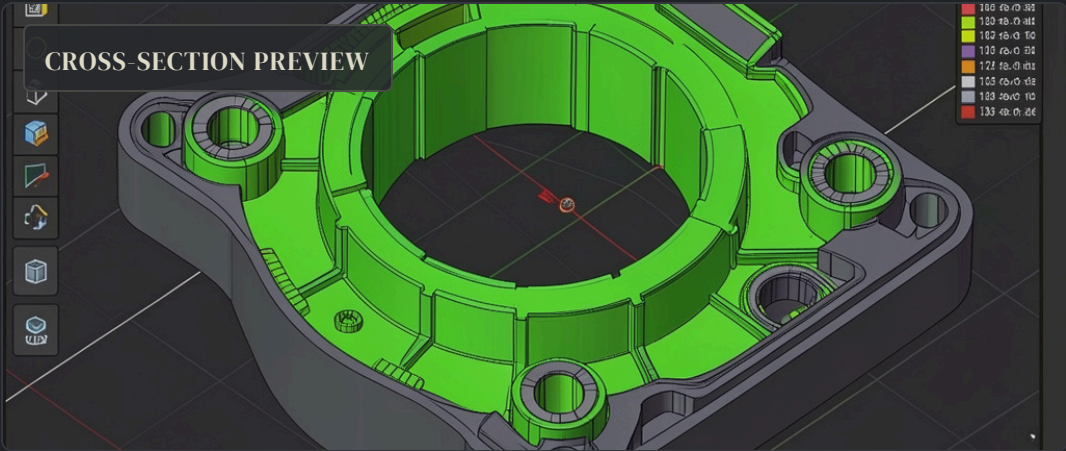


15% Grid Infill

A 15% grid leaves air pockets that can collapse around fasteners.

Construction Type

Sparse Internal Grid



Recommended structural settings:

Infill

100%

Pattern

Rectilinear

Wall loops

10

Top/bottom layers

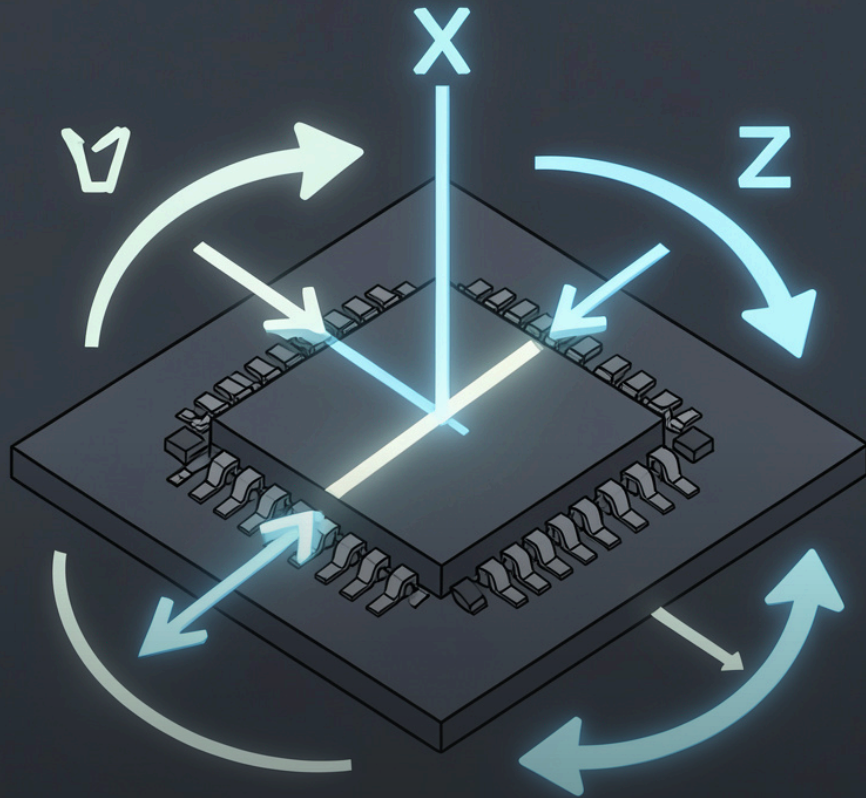
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Continuous carbon-reinforced material paths better support stress-bearing regions.

IMU Sensing: Raw Accelerometer & Gyroscope Ingestion

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INERTIAL MEASUREMENT UNIT · 6-DOF PROCESSING PIPELINE



SENSOR COORDINATE FRAME

MPU6050 3-Axis Gyroscope & 3-Axis Accelerometer Orientation Diagram

- The MPU6050 supplies angular rate from its gyroscope and acceleration from its accelerometer.
- Configured ranges are $\pm 500^\circ/\text{s}$ for the gyro and $\pm 8\text{g}$ for acceleration.
- Fast I2C operates at 400 kHz for timely sensor access.

INGESTION & CONVERSION EQUATIONS

Gyroscope conversion:

$$\text{Rate}_{\text{gyro}} = \text{RawGyro} / 65.5$$

Accelerometer tilt estimate:

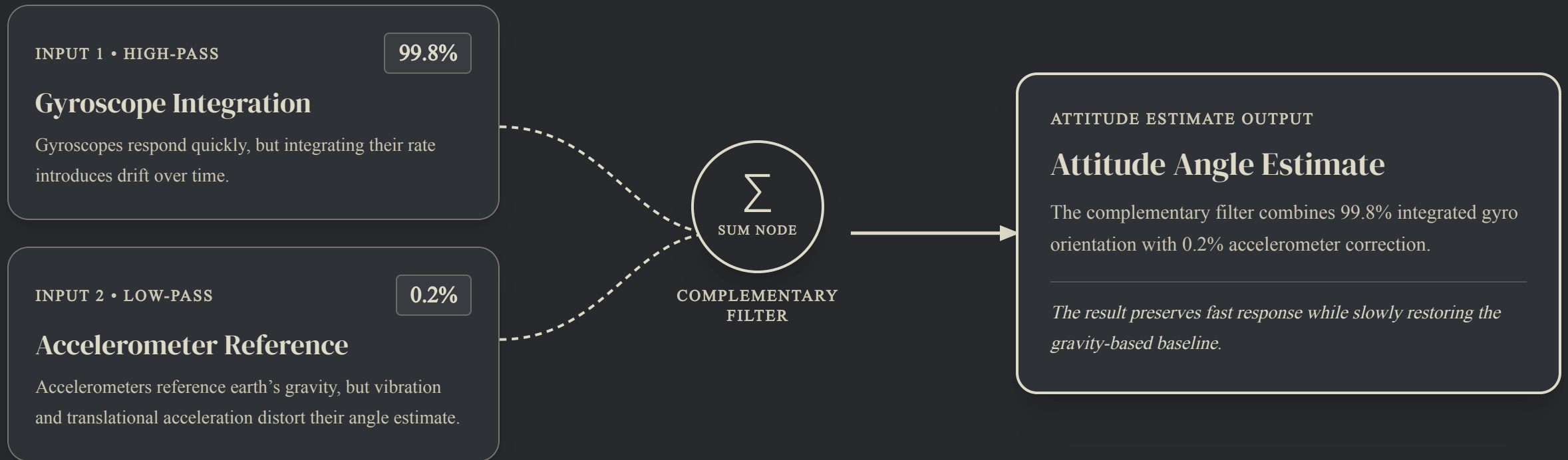
$$\text{Angle}_{\text{accel}} = \text{atan2}(A_y, A_z) \times 180/\pi$$

C++ Ingestion Snippet

mpu6050_read.cpp

```
AccX = rawAccX / 4096.0;  
RawRatePitch = rawGyroX / 65.5;
```

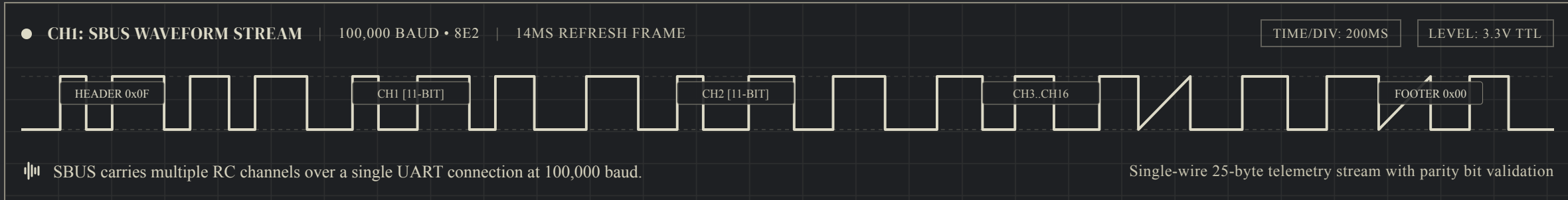
Sensor Fusion via Complementary Filtering



```
C++ angle = 0.998 * (angle + Rate * dt) + 0.002 * AngleAccel;
```

Real-time attitude orientation fusion step

Radio Signal Ingestion: Non-Inverted SBUS



Signal Transformation Pipeline

SEQUENTIAL INGESTION STAGES

STEP 01

01

Raw Channel Range

282 – 1722 ticks

Raw channel values from 282 to 1722 are converted into familiar 1000–2000 μ s servo-control bounds.

STEP 02

02

Formula Mapping

Linear Interpolation

Formula:
 $PWM_{\mu s} = \text{map}(SBUS_{raw}, 282, 1722, 1000, 2000)$

STEP 03

03

Receiver Layer

Control Subsystem

The receiver layer updates channel values before deadband, trim, throttle, and control calculations.

- Deadband Filtering
- Trim Offsets
- Throttle Safety
- Flight Dynamics

16-CH Ingestion

CONVERTED SERVO RANGE



1000–2000 μ s Standard RC PWM Output

Min: 1000 μ s

Neutral: 1500 μ s

Max: 2000 μ s

IMPLEMENTATION LOGIC

C++ FIRMWARE READ LOOP

```
if (sbus_rx.Read()) {
    ReceiverValue[i] = map(sbus_data.ch[i], 282, 1722, 1000, 2000)
}
```

- Synchronous packet ingestion per telemetry frame cycle

Signal Conditioning: Filtering Stick Noise

● **HARDWARE FLUCTUATION**

Transmitter potentiometers and spring mechanisms can introduce approximately $\pm 10\ \mu\text{s}$ of center noise.

Without conditioning, the controller interprets micro-jitters as intentional commands and continually adjusts the motors.

● **15 MS DEADBAND RULE**

Stable Center

If $|\text{StickVal} - 1500| < 15$, set $\text{StickVal} = 1500$

A 15 μs deadband establishes a stable center, eliminating erratic motor oscillation while keeping response sharp.

SCOPE

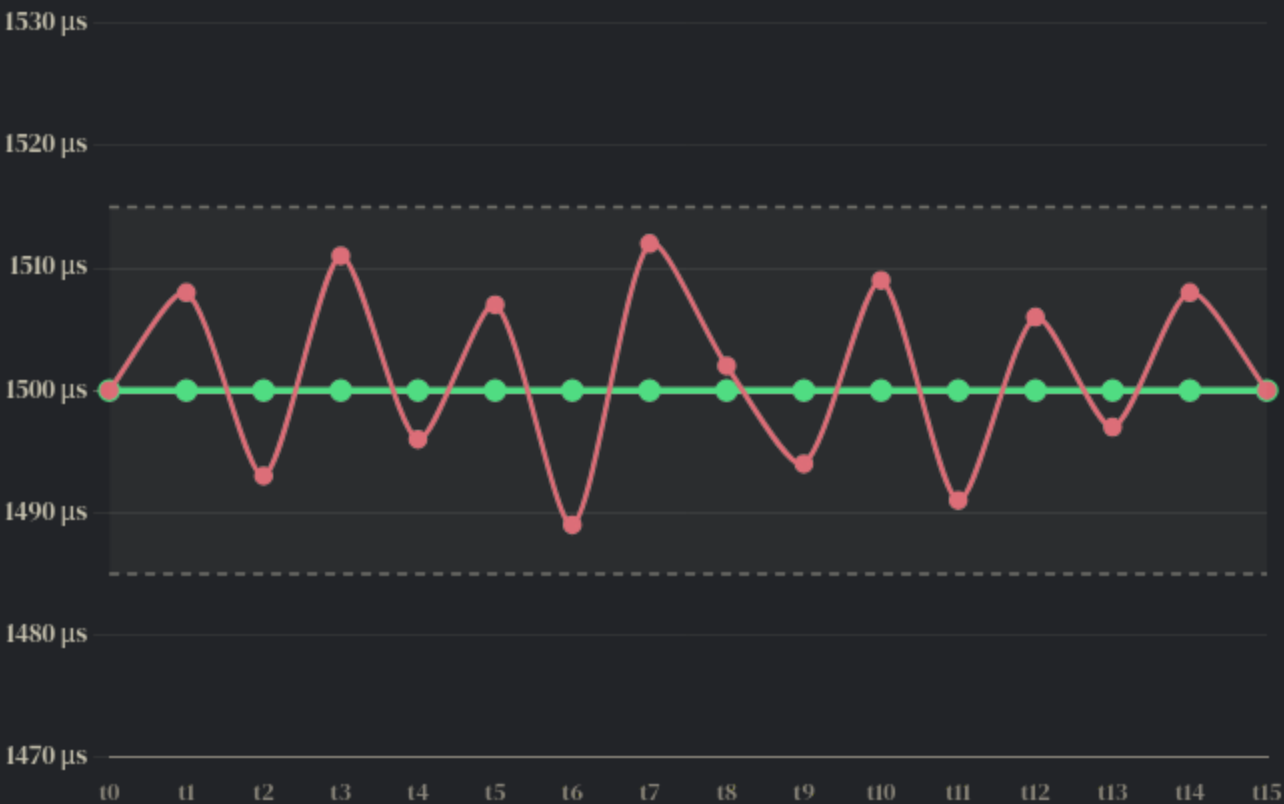
Roll, pitch, and yaw channels receive the same center-locking treatment.

— Noisy Input Signal (Raw) — Filtered Output (1500 μs Locked)

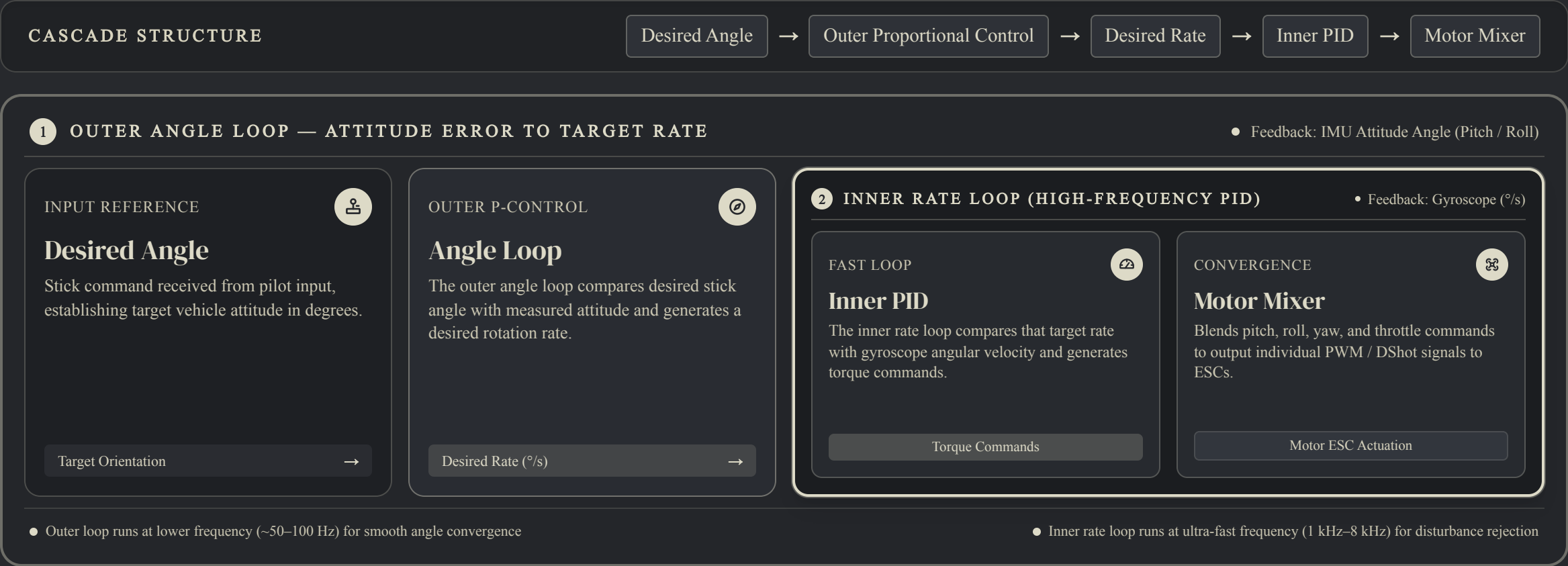
--- $\pm 15\ \mu\text{s}$ Deadband Thresholds

Pulse Width Modulation (μs) vs. Sample Time

Target Neutral: 1500 μs

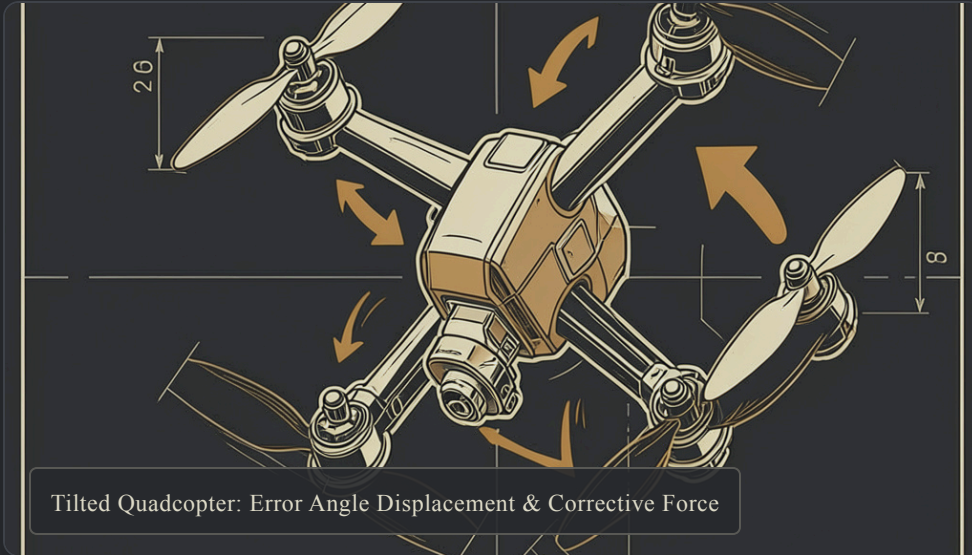


Control Theory: Cascaded Angle-to-Rate PID Loops



PID Component 1: Proportional Gain (P)

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Step Response Curve Across Varying Proportional Gain ● Low Gain ● Optimal (Tuned) ● Excessive Gain



Formula: $P_{term} = K_p \times e(t)$.

Direct linear error proportional relationship

ATTITUDE CORRECTION LOGIC

cpp

```
ErrorAnglePitch = DesiredAnglePitch - complementaryAnglePitch;
```

```
PtermPitchAngle = PAnglePitch * ErrorAnglePitch;
```

Proportional response calculated per control cycle

- Proportional control supplies immediate corrective effort based on the present error.
- A larger gain produces stronger and faster correction when the aircraft is displaced.
- Excessive gain creates high-frequency oscillation because the controller repeatedly overcorrects around the desired attitude.

Integral Gain (I)

Integral control stores error over time and counters persistent offsets.

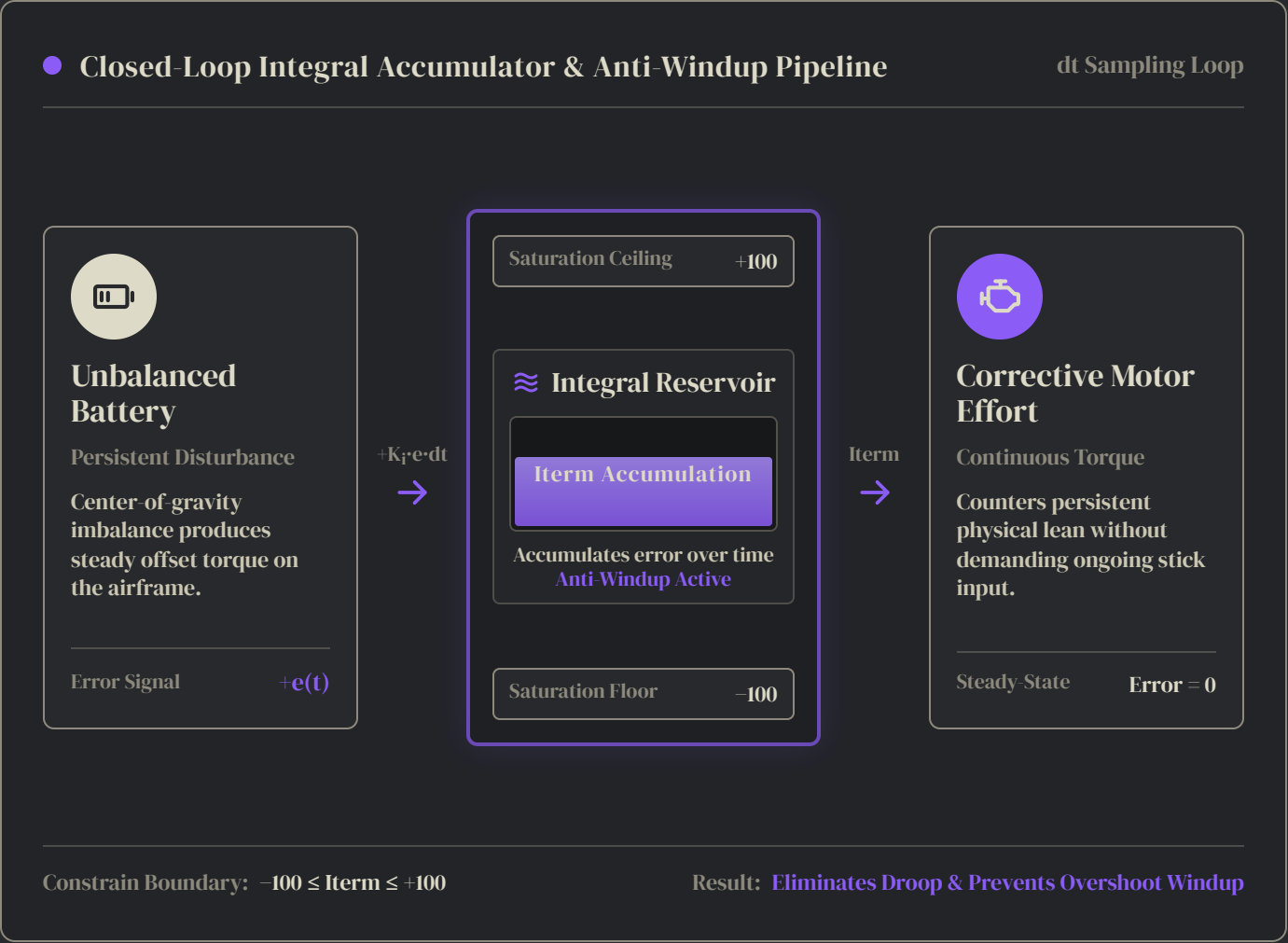
It compensates for effects such as an unbalanced battery or a frame that requires continuous corrective torque.

Approximation: $I_{term} = I_{prev} + K_i \times e(t) \times dt$

Anti-windup limits constrain the stored value to -100 through $+100$, preventing unlimited accumulation during saturation.

ALGORITHM IMPLEMENTATION C++

```
Iterm = PrevIterm + IRate * ErrorRate * dt;  
Iterm = constrain(Iterm, -100, 100);
```



PID Component 3: Derivative Gain (D)

Derivative control responds to how quickly error changes rather than only its size.

MATHEMATICAL REPRESENTATION

Formula: $D_{\text{term}} \approx K_d \times [e(t) - e(t-1)] / dt$

It acts as damping, reducing overshoot and smoothing the arrival at level hover.

Excessive derivative sensitivity can amplify noisy measurements, so gyro quality and timing consistency matter.

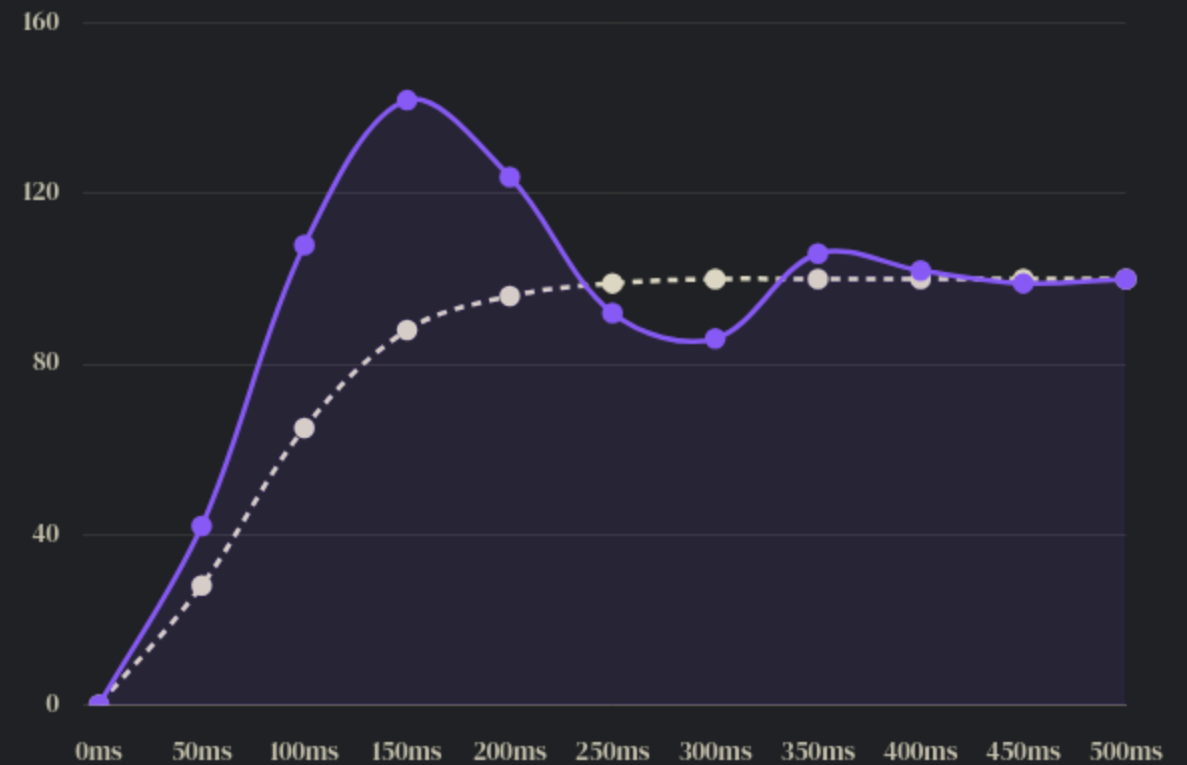
C++ IMPLEMENTATION

Loop Update

```
Dterm = DRate * ((ErrorRate - PrevErrorRate) / dt);  
Input = Pterm + Iterm + Dterm;
```

Step Response Comparison

Settling Time vs Damping



— Undamped / Underdamped (High Overshoot)

— Critically Damped (Tuned D)

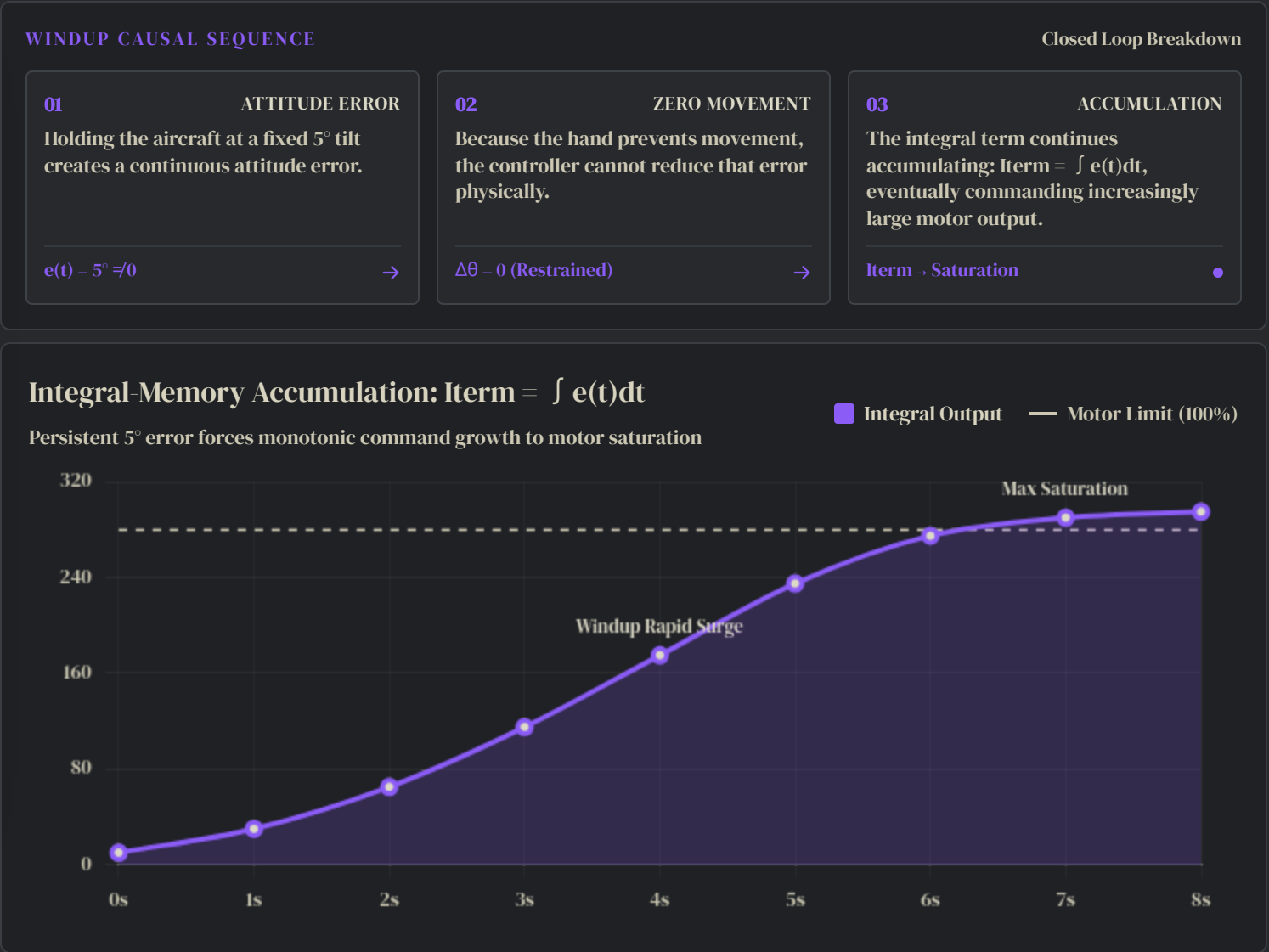
Control Theory Paradox: The Hand-Test Trap

● FIXED 5° TILT TEST CONSTRAINT



Artificially restraining frame rotation creates an unresolvable feedback loop.

The apparent “lack of power” is actually windup caused by an artificial test constraint, not a flight-control failure.



• Safety Strategy: Pure PD Tuning Mode

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Bench & Hand Alignment Safety

Alignment tests should disable integral memory before motors are checked in hand.



Dynamic Response Balance

Pure PD mode retains proportional correction and derivative damping without allowing error accumulation.

● ANTI-WINDUP SAFEGUARD

If movement is physically prevented, the controller cannot build an integral command against the operator's grip.

Operating Mode Selector

STATE ARCHITECTURE

SAFE TESTING



Pure PD Mode

Hand-alignment safety enabled

IN-FLIGHT ONLY

Integral Active

Full PID Flight Control

Accumulates error under resistance

Configuration

Pure PD Safe-Testing Parameters

P

1.8

PRate

Proportional Gain

D

0.012

DRate

Derivative Damping

I

0.0

IRate

Disabled (Zero Acc.)

Memory Protection: Anti-Windup Clamping

- Integral action is useful in flight because it counters battery imbalance and persistent frame torque.

- The stored value must remain bounded during output saturation or constrained tests.

SATURATION FUNCTION

Constraint: $I_{clamped} = \max(-100, \min(100, I_{raw}))$

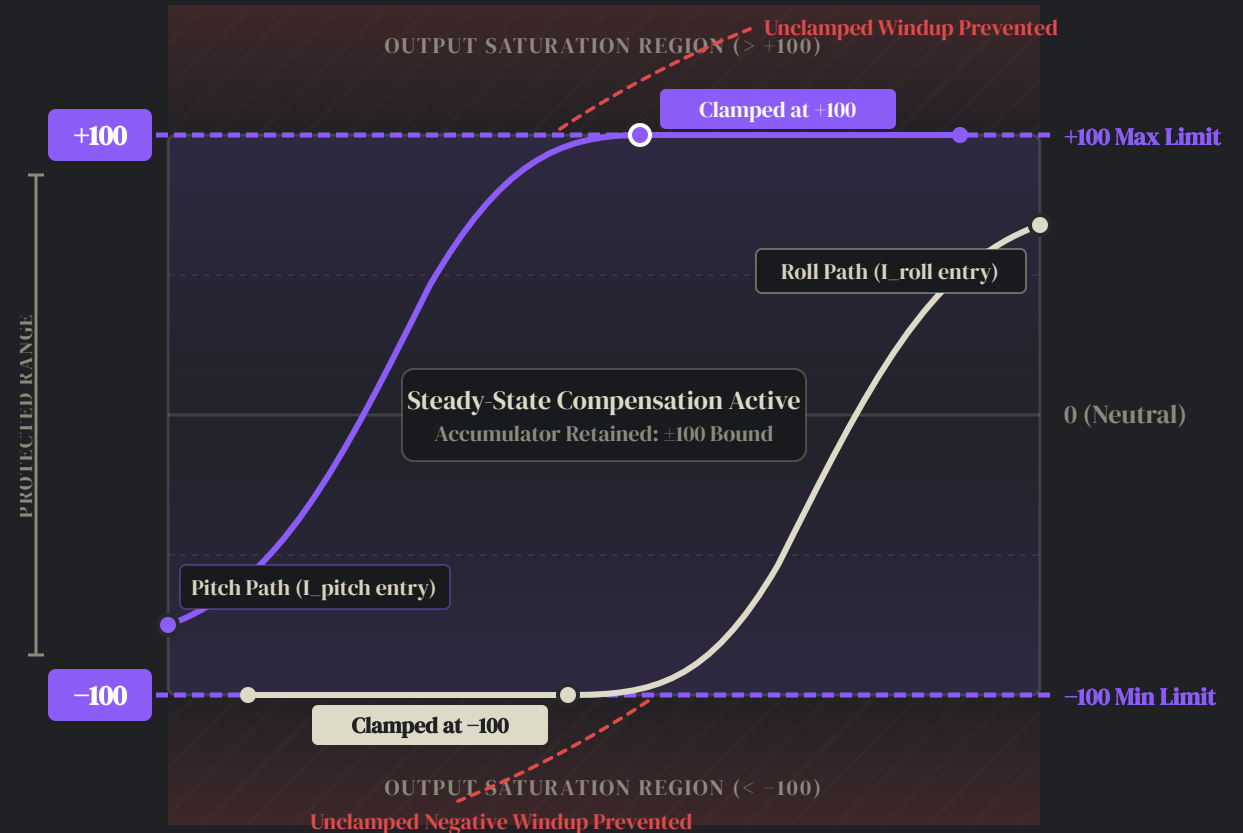
- Applying this limit to pitch and roll integral accumulators keeps stored corrective effort within a manageable range while preserving steady-state compensation.

— Pitch Channel (I_{pitch}) — Roll Channel (I_{roll})

Bounded Memory Accumulator

Dual-axis saturation guard & anti-windup clamping threshold

Limit: ± 100 Units



Fail-Safe Architecture: *Instant Memory Wipe*

The CH5 disarm switch provides an immediate stop condition.

The routine combines output shutdown with state reset, rather than relying on motor commands alone.

LOGICAL TRANSITION RULE

$CH5 < 1500 \rightarrow \text{Motors} = 900 \mu\text{s};$
 $\text{PrevError} = 0; \text{PrevIterm} = 0$

1

PHYSICAL BRANCH

Motor Output Shutdown

When $CH5 < 1500$, every motor command is forced to $900 \mu\text{s}$.

- Forces throttle output to $900 \mu\text{s}$ to ensure instant, total propulsion cutoff.
- Overrides all active flight-mode commands and ESC mixer outputs unconditionally.

FORCED OUTPUT VALUE

$900 \mu\text{s}$

2

STATE BUFFER BRANCH

Memory & PID Reset

Resetting previous errors and integral terms prevents old control memory from reappearing after rearming.

- Clears PID history completely so integral accumulation does not cause windup.
- Guarantees a clean, uncorrupted baseline state upon subsequent arming switch transition.

CLEARED REGISTERS

$\text{PrevError} = 0 \cdot \text{PrevIterm} = 0$

Hyper-Precise Controls: Cubic Trim Mapping

CH9 & CH10 Dual-Axis Attitude Trimming (±3.0° Boundary Limits)

CONTROL DYNAMICS

Linear trim mapping can make center-stick adjustments coarse and twitchy.

Cubic mapping uses x^3 , reducing sensitivity near center while preserving useful travel near the endpoints.

MATHEMATICAL FORMULATION

$$\text{Trim}_{\text{angle}} = \text{BaseOffset} + [((\text{RawInput} - 1500) / 500)^3 \times 3.0^\circ]$$

Center at 1500 μs (0.0°) • Endpoints at 1000 μs & 2000 μs (±3.0°)

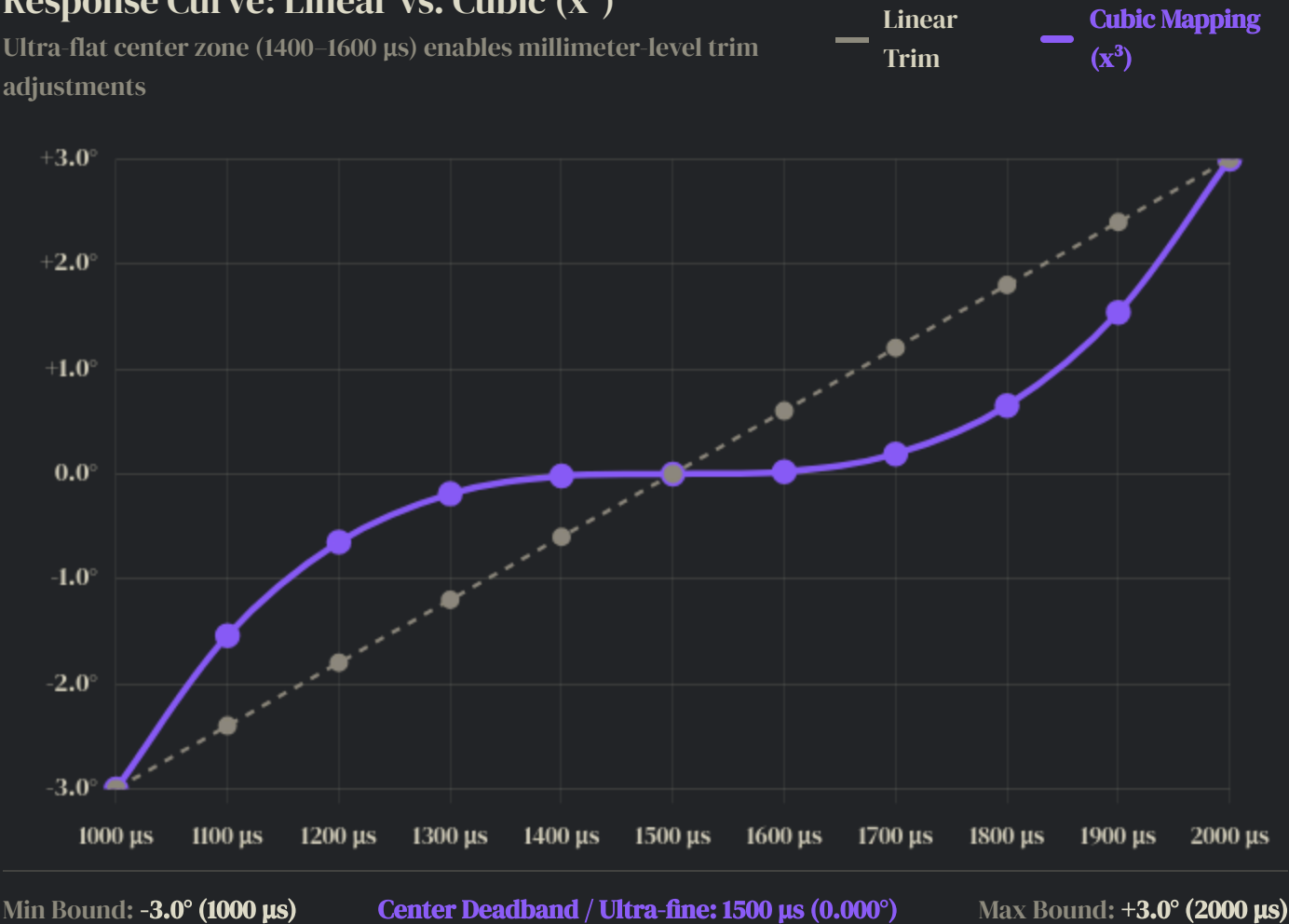
HARDWARE CHANNEL MAPPING

CH9 and CH10 provide roll and pitch trim with a strict maximum adjustment of ±3.0°.

The method supports fine hover leveling without sacrificing full slider range.

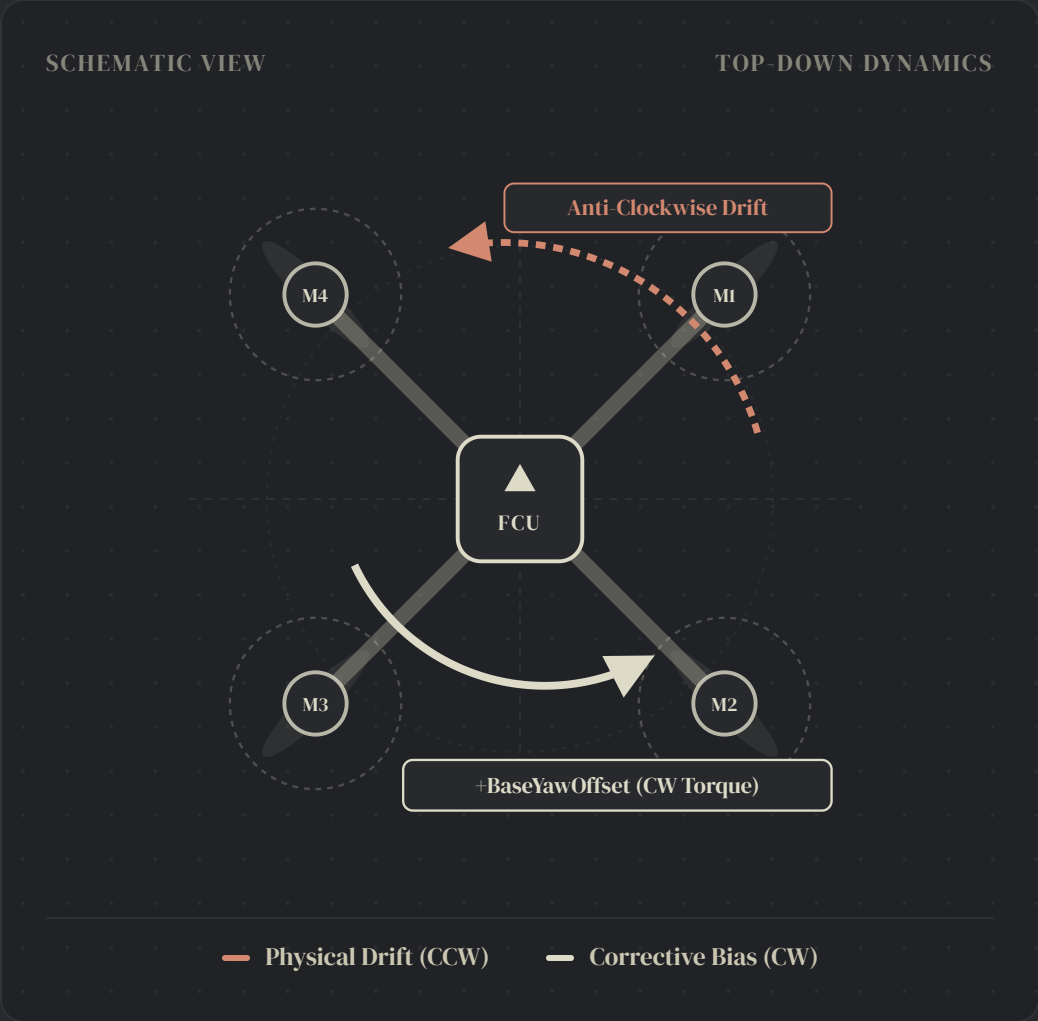
Response Curve: Linear vs. Cubic (x^3)

Ultra-flat center zone (1400–1600 μs) enables millimeter-level trim adjustments



Yaw Control

Injecting Software Yaw Bias



01 / PHYSICAL IMBALANCE

Small frame twists or unequal motor effects can create persistent rotational drift.

02 / SOFTWARE COMPENSATION

A constant `BaseYawOffset` adds corrective demand to the desired yaw-rate calculation.

03 / RATE CALCULATION

FORMULA

$$DesiredRateYaw = 0.15 \times (YawStick - 1500) + BaseYawOffset$$

04 / CALIBRATION IMPACT

Example: `BaseYawOffset = 5.0` supplies a steady correction against anti-clockwise drift, reducing the need for continuous manual yaw input.

Dynamic Throttle Expo Curve

Linear throttle response can make small stick changes produce abrupt altitude increases around hover.

- The algorithm reduces sensitivity near the 1500 μs hover region and changes responsiveness farther from center.
- Above 1520 μs or below 1480 μs, throttle difference determines dynamic sensitivity.

DYNAMIC TRANSFER MODEL

Hover Window ($|T - 1500| \leq 20 \mu s$)

$$R(T) = T_0 + k_{fine} \cdot \Delta T$$

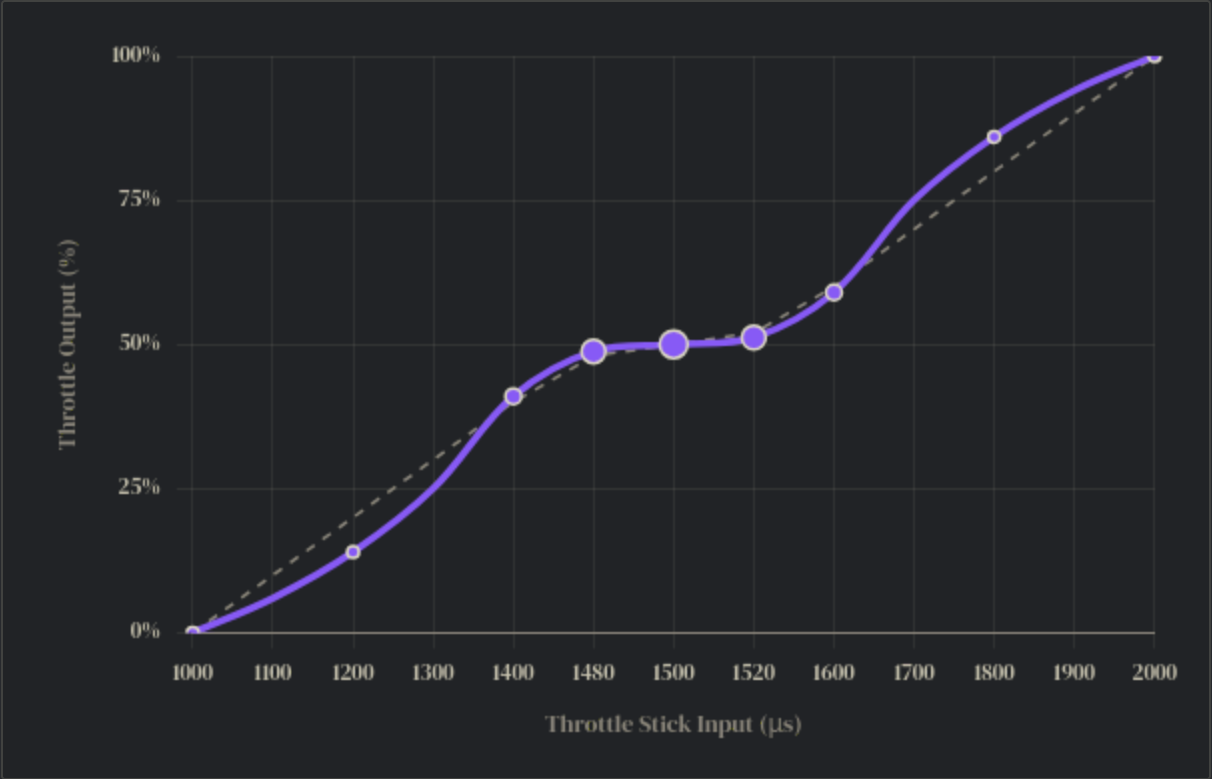
Dynamic Authority ($|\Delta T| > 20 \mu s$)

$$R(T) = T_0 + \text{sgn}(\Delta T) \cdot [\delta_0 + \alpha|\Delta T|^{\gamma}]$$

The resulting curve provides finer altitude control during tuning while retaining stronger response for larger commands.

Throttle-Input Response Curve

Center Hover: 1500 μs

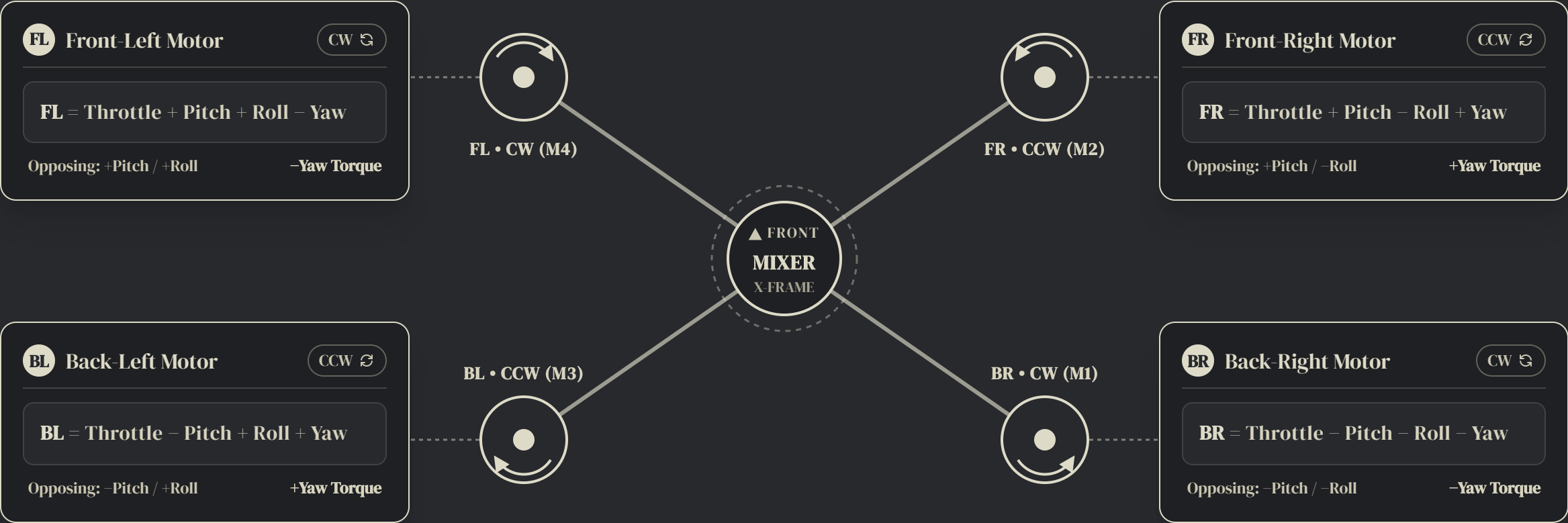


Dynamic Expo Curve Standard Linear

1480–1520 μs: Reduced Hover Sensitivity

Torque & Thrust Mixing: X-Frame Motor Matrix

- The mixer combines throttle, pitch, roll, and yaw commands into four motor outputs.
- Pitch and roll modify opposing motor pairs; yaw modifies diagonal pairs according to counter-rotating propeller torque.

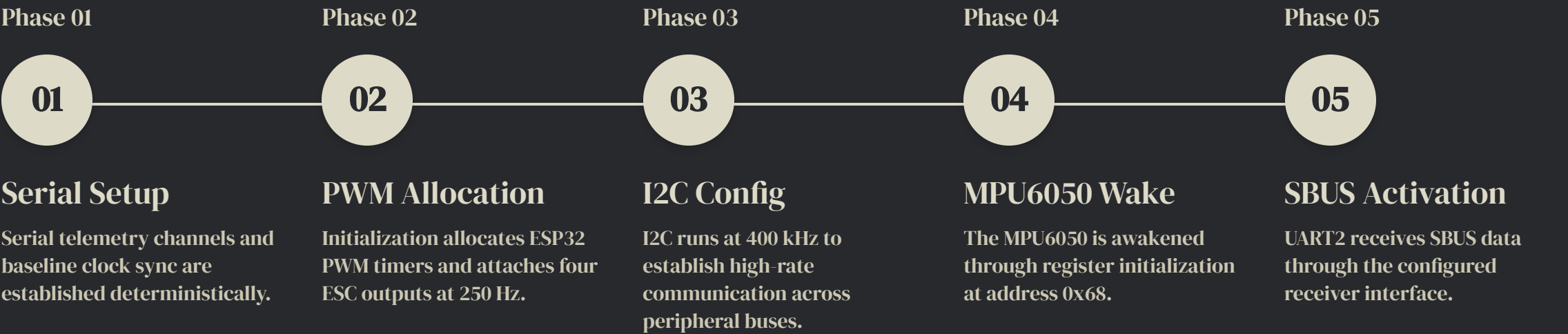


• **ESC Output Constraint:** Each result is constrained before being sent to the ESC.

MIN / MAX PULSE BOUND [0% – 100%]

Firmware Core: Setup & Ingestion Engine (V69.4)

The setup phase establishes deterministic peripherals before the main loop begins sensor ingestion and attitude estimation.



Firmware Core: PID Execution & Control Loop (V69.4)



Pre-Flight Playbook & Engineering Checklist

